

SketchUp Move Tool

100% Accurate Technical Specification

For Nestup 3D Cabinet Visualizer Implementation

All values researched from official SketchUp documentation & Ruby API

1. Exact Visual Specifications

1.1 Inference Point Indicators

Type	Shape	Color	Hex	Size	ScreenTip Text
Endpoint	Circle	Green	#00FF00	6px	"Endpoint"
Midpoint	Circle	Cyan	#00FFFF	6px	"Midpoint"
On Edge	Square	Red	#FF0000	6px	"On Edge"
On Face	Diamond	Dark Blue	#0000CC	6px	"On Face"
Intersection	X mark	Red	#FF0000	6px	"Intersection"
Origin	Circle	Yellow	#FFFF00	6px	"Origin"
Center	Circle	Green	#00FF00	6px	"Center"
In Group	(same)	Magenta	#FF00FF	6px	"[type] in Group"
In Component	(same)	Magenta	#FF00FF	6px	"[type] in Component"

Source: *SketchUp Ruby API View.draw_points() documentation* - default size = 6 pixels

1.2 Point Indicator Shapes (Ruby API Style Constants)

Style ID	Shape	Use For	Visual
1	Open Square	On Edge	□ (hollow)
2	Filled Square	Alternative	■ (solid)
3	Plus "+"	Default style	+ (crosshair)
4	Cross "X"	Intersection	× (diagonal)
5	Star "*"	Alternative	* (6-point)
6	Open Triangle	Alternative	△ (hollow)
7	Filled Triangle	Alternative	▲ (solid)

1.3 Line Styles

State	Pattern	Width	Stipple	Visual
Inference Suggested	Dashed	1px	"_"	- - - - -
Inference LOCKED	Solid	2-3px	""	————(bold)
Bounding Box	Dashed	1px	"."	 (dotted)

Line Stipple Patterns (Ruby API):

- "." = Dotted Line
- "-" = Short Dashes
- "_" = Long Dashes
- "-." = Dash-Dot-Dash
- "" (empty) = Solid Line

2. ScreenTip (Tooltip) Behavior

Property	Value	Notes
Display Delay	~0ms (instant)	Appears immediately on hover
Auto-hide Time	3-5 seconds	Disappears after this duration
Position	Below cursor	Small yellow/cream background box
Font	System default	Small sans-serif, ~11px
Background	Light yellow	Standard Windows/Mac tooltip style

Exact ScreenTip Text Formats:

- "Endpoint" - vertex at end of edge
- "Midpoint" - center of edge
- "On Edge" - point along edge
- "On Face" - point on surface
- "Intersection" - where edges cross
- "Center" - center of arc/circle
- "Origin" - model origin point
- "From Point" - linear inference from point
- "On Red Axis" / "On Green Axis" / "On Blue Axis"
- "Constrained on Line from Point" - locked linear
- "Endpoint in Group" / "Midpoint in Component" - nested geometry

3. Snap Radius & Detection

Parameter	Value	Notes
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Snap Radius	~7-10 screen pixels	NOT configurable by user
Geometric Tolerance	0.001 inches	Internal precision limit
Cannot be disabled	Always active	Core SketchUp feature

Inference Priority Order (highest to lowest):

1. **Endpoint** - vertex/corner points
2. **Midpoint** - center of edges
3. **Intersection** - where edges/faces cross
4. **Center** - arc/circle centers
5. **On Edge** - arbitrary edge point
6. **On Face** - surface point
7. **Guide Point** - construction geometry

Priority Conflict: When "Constrained on Line from Point" conflicts with "Endpoint", Endpoint SHOULD win but sometimes doesn't - this is a known SketchUp quirk.

4. Bounding Box Grips (SketchUp 2020+)

Grip Type	Count	Normal	Obscured	Notes
Corners	8	Gray #808080	Blue #0066FF	Default visible
Edge Midpoints	12	Gray #808080	Blue #0066FF	Cycle with CMD/ALT
Face Centers	6	Gray #808080	Blue #0066FF	Cycle with CMD/ALT
Object Center	1	Gray #808080	Blue #0066FF	Cycle with CMD/ALT

Grip Visual Properties:

- **Shape:** Small filled circle (dot)
- **Size:** ~6 pixels diameter (screen-space, doesn't scale with zoom)
- **Obscured Behavior:** When corner is behind object, grip turns blue and object becomes temporarily transparent

Cycling Order (CMD on macOS / ALT on Windows):

8. Corners (8 points) ← DEFAULT
9. Edge Midpoints (12 points)
10. Face Centers (6 points)
11. Object Center (1 point)

5. Axis Locking (Arrow Keys)

CRITICAL: Arrow keys are TOGGLES. Press once = lock. Press same key again = unlock. They do NOT need to be held down.

Key	Axis	Color	Hex	Behavior
→ Right	X-axis (Red)	Red	#FF0000	Toggle lock horizontal left-right
← Left	Y-axis (Green)	Green	#00FF00	Toggle lock horizontal front-back
↑ Up	Z-axis (Blue)	Blue	#0000FF	Toggle lock vertical
↓ Down	Parallel/Perp	Magenta	#FF00FF	3-state: Parallel → Perpendicular → Off

Down Arrow 3-State Toggle (relative to hovered edge):

12. First press: Lock PARALLEL (magenta solid line)
13. Second press: Lock PERPENDICULAR (magenta solid line)
14. Third press: UNLOCK (back to free movement)

Shift Key: Hold Shift to TEMPORARILY lock current inference. Releasing Shift unlocks it. This is different from arrow key toggles.

Visual Feedback:

- **Suggesting (not locked):** Dashed line in axis color
- **LOCKED:** Solid BOLD line in axis color
- **ScreenTip shows:** "On Red Axis" becomes "Constrained on Line" when locked

6. Cursor Appearances

State	Cursor	Description
Move Tool Active	Four black arrows (cross/move icon)	↑←↓→ pointing outward from center
Copy Mode (Ctrl/Opt)	Four arrows + small "+" sign	Plus appears at cursor corner
Stamp Mode (Ctrl×2)	Four arrows + stamp indicator	Double-press Ctrl/Option
Hovering Snap Point	Four arrows + inference indicator	Colored shape appears at point
Moving (active)	Four arrows (moving)	Preview geometry follows cursor
Axis Locked	Four arrows + lock indicator	Shows which axis is locked

7. Autofold Behavior

When moving loose geometry (not grouped) that would bend a face, SketchUp automatically creates fold lines to keep all faces planar.

Activation:

- **Alt key (Windows) / Command key (macOS):** Force Autofold mode
- Allows moving point/edge/face in ANY direction, creating fold edges as needed

Behavior:

- Faces MUST remain planar in SketchUp
- If stretching would bend a face, new edges are created to maintain planarity
- Sometimes SketchUp limits operation instead of creating fold lines

8. VCB (Measurements Box) Input

CRITICAL: NEVER click in the VCB. Just start typing. It always has focus during tool operations.

Format	Example	Result
Distance	500mm, 20cm, 1.5m	Move this distance along current direction
Imperial	20', 6", 3' 6"	Feet/inches, space-separated compound
Negative	-500mm, -2'	Move in OPPOSITE direction
Global Coords	[3', 4', 5']	Square brackets = absolute from origin
Relative Coords	<3', 4', 5'>	Angle brackets = relative to start point
Array Multiply	x5 or *5	Create 5 copies at same spacing
Array Divide	/5	Divide total distance into 5 equal parts

Re-typing: You can keep typing new values until you click elsewhere or switch tools. Each new value replaces the previous.

9. Copy Mode & Arrays

Activation:

- **Ctrl (Windows) / Option (macOS):** TAP once to toggle copy mode. + appears at cursor.
- **TAP twice:** Enter STAMP mode. Each click creates another copy.

Array Modifiers (type IMMEDIATELY after completing copy):

- **x5 or *5:** Creates 5 TOTAL copies at the same spacing interval
- **/5:** Divides the distance, creating 4 copies evenly distributed (5 spaces, 6 items total)
- **Re-typeable:** You can keep adjusting until tool change

10. Move Tool State Machine

State	Description	Entry Action	Exit To
IDLE	Tool active, awaiting click	Press M key	Click → PICKING
PICKING	First click done, start point set	Click geometry/point	Move mouse → MOVING
MOVING	Preview following cursor	Mouse movement	Click/Enter → COMPLETE
COMPLETE	Move done, VCB still active	Second click or Enter	New value → IDLE

11. Three.js Implementation Reference

11.1 Constants (exact values)

```
// EXACT COLORS (from SketchUp documentation) const INFERENCE_COLORS = {   endpoint: 0x00FF00,
// Green   midpoint: 0x00FFFF,       // Cyan   onEdge: 0xFF0000,       // Red   onFace:
0x0000CC,       // Dark Blue   intersection: 0xFF0000, // Red   origin: 0xFFFF00,       //
Yellow   center: 0x00FF00,       // Green   inGroup: 0xFF00FF,       // Magenta (overrides
all others) }; const AXIS_COLORS = {   x: 0xFF0000,       // Red   y: 0x00FF00,
// Green   z: 0x0000FF,       // Blue   parallel: 0xFF00FF,       // Magenta
perpendicular: 0xFF00FF, // Magenta }; const GRIP_COLORS = {   normal: 0x808080,       //
Gray   obscured: 0x0066FF,       // Blue   hover: 0x0066FF,       // Blue }; // EXACT SIZES
(from Ruby API draw_points) const INDICATOR_SIZE = 6; // pixels (default in SketchUp) const
SNAP_RADIUS = 10; // pixels (approximate) const LINE_WIDTH_NORMAL = 1; const
LINE_WIDTH_LOCKED = 3; // LINE STIPPLE PATTERNS const LINE_STIPPLE = {   dotted: ".",
// Bounding box   dashed: "-",       // Inference suggesting   solid: "",       //
Inference locked };
```

11.2 ScreenTip Implementation

```
class ScreenTip {   constructor() {     this.visible = false;     this.text = "";
this.hideTimer = null;     this.DISPLAY_DURATION = 4000; // 3-5 seconds     // Create DOM
element     this.element = document.createElement('div');     this.element.style.cssText = `
position: absolute;     background: #FFFFCC;     border: 1px solid #808080;     padding:
2px 6px;     font: 11px sans-serif;     color: #000;     pointer-events: none;
display: none;     z-index: 1000;     `;     document.body.appendChild(this.element);   }
show(text, x, y) {     this.text = text;     this.element.textContent = text;
this.element.style.left = (x + 15) + 'px';     this.element.style.top = (y + 20) + 'px';
this.element.style.display = 'block';     // Auto-hide after duration
clearTimeout(this.hideTimer);     this.hideTimer = setTimeout(() => this.hide(),
this.DISPLAY_DURATION);   }   hide() {     this.element.style.display = 'none';   }
getTooltipText(inferenceType, isInGroup = false, isInComponent = false) {     const base = {
endpoint: "Endpoint",     midpoint: "Midpoint",     onEdge: "On Edge",     onFace: "On
Face",     intersection: "Intersection",     center: "Center",     origin: "Origin",
}[inferenceType] || "";     if (isInComponent) return base + " in Component";     if
(isInGroup) return base + " in Group";     return base;   } }
```

11.3 Axis Lock Manager (Exact Behavior)

```
class AxisLockManager {   constructor() {     this.lockedAxis = null;     // 'x' | 'y' |
'z' | null     this.parallelPerpState = 0;     // 0=none, 1=parallel, 2=perpendicular
this.referenceEdgeDirection = null;   }   handleArrowKey(key, hoveredEdgeDirection = null) {
switch(key) {     case 'ArrowRight':       // TOGGLE: if already X, unlock; else lock X
this.lockedAxis = (this.lockedAxis === 'x') ? null : 'x';       this.parallelPerpState = 0;
break;     case 'ArrowLeft':       this.lockedAxis = (this.lockedAxis === 'y') ?
null : 'y';       this.parallelPerpState = 0;       break;     case 'ArrowUp':
this.lockedAxis = (this.lockedAxis === 'z') ? null : 'z';       this.parallelPerpState = 0;
break;     case 'ArrowDown':       // 3-STATE TOGGLE: parallel → perpendicular →
off       this.lockedAxis = null;       this.parallelPerpState = (this.parallelPerpState +
1) % 3;       if (this.parallelPerpState > 0) {         this.referenceEdgeDirection =
hoveredEdgeDirection;       }       break;   }   return this.getState(); }
getState() {   return {     lockedAxis: this.lockedAxis,     parallelPerpState:
```



```

this.parallelPerpState,      isLocked: this.lockedAxis !== null || this.parallelPerpState >
0,      lineStyle: this.isLocked() ? 'solid' : 'dashed',      lineWidth: this.isLocked() ? 3
: 1,      };    }    isLocked() {      return this.lockedAxis !== null || this.parallelPerpState
> 0;    }    reset() {      this.lockedAxis = null;      this.parallelPerpState = 0;
this.referenceEdgeDirection = null;    } }

```

12. Testing Checklist

Visual Indicators:

- ☐ Green circle (6px) appears at endpoints
- ☐ Cyan circle (6px) appears at midpoints
- ☐ Red square (6px) appears on edges
- ☐ Dark blue diamond (6px) appears on faces
- ☐ All indicators turn magenta inside groups/components
- ☐ ScreenTip appears with correct text
- ☐ ScreenTip disappears after 3-5 seconds

Axis Locking:

- ☐ Arrow keys TOGGLE (not hold)
- ☐ Right arrow = X (red)
- ☐ Left arrow = Y (green)
- ☐ Up arrow = Z (blue)
- ☐ Down arrow cycles: parallel → perpendicular → off
- ☐ Dashed line when suggesting
- ☐ Solid bold line when LOCKED

Bounding Box Grips:

- ☐ Gray dots at corners by default
- ☐ CMD/ALT cycles grip types
- ☐ Blue when obscured/hovered
- ☐ Object becomes transparent when showing obscured grip

VCB & Copy:

- ☐ Distance input works (500mm, 3' 6")
- ☐ Negative distance moves opposite
- ☐ [x,y,z] global coordinates work
- ☐ <x,y,z> relative coordinates work
- ☐ Ctrl/Option shows + at cursor
- ☐ x5 creates 5 copies
- ☐ /5 divides into 5 equal parts

13. Summary of Exact Values

Parameter	Exact Value
Indicator size (default)	6 pixels
Snap radius	~7-10 pixels (not configurable)
Geometric tolerance	0.001 inches
ScreenTip display time	3-5 seconds
Line width (normal)	1 pixel
Line width (locked)	2-3 pixels (bold)
Endpoint color	#00FF00 (green)
Midpoint color	#00FFFF (cyan)
On Edge color	#FF0000 (red)
On Face color	#0000CC (dark blue)
In Group/Component	#FF00FF (magenta - overrides all)
Grip normal color	#808080 (gray)
Grip obscured color	#0066FF (blue)

Document compiled from: Official SketchUp Help documentation, SketchUp Ruby API (View.draw_points, View.line_stipple, View.line_width), SketchUp Community Forums, MasterSketchUp.com tutorials, LinkedIn Learning courses, and SketchUp Quick Reference Cards (2025/2026).